Experiment - 1

Compilation Stages of a C Program

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1. Writing the C Program

```
UW PICO 5.09

#include<stdio.h>
#define PI 3.14

int main(void)
{
    int radius;
    float area;

    printf("Enter the radius: ");
    scanf("%d", &radius);

    area = PI * radius * radius;
    printf("Area = %f\n", area);

    return 0;
}
```

The file radius.c is open in the Pico editor. It includes the standard input/output header, reads an integer radius, calculates the area using 3.14 x radius x radius, and prints the result.

2. Compiling and Running

The terminal shows the program being compiled and executed. Radius 12 is entered, and the program prints Area = 452.160004, confirming that it runs successfully.

```
Last login: Tue Aug 18 00:06:12 on ttys000
bhavya@Bhavyas-MacBook-Air ~ % nano radius.c
bhavya@Bhavyas-MacBook-Air ~ % clang radius.c
bhavya@Bhavyas-MacBook-Air ~ % ./radius
Enter the radius:12
Area = 452.160004
bhavya@Bhavyas-MacBook-Air ~ %
```

3. Viewing the Preprocessed Program

```
# 472 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/_stdio.h" 3 4
extern const int sys_nerr;
extern const char *const sys_errlist[];

int asprintf(char * *restrict, const char * restrict, ...) __attribute__((__format__ (__printf__, 2, 3)));
char * ctermid_r(char *);
char * fgetln(FILE *, size_t *__len);
const char *fmtcheck(const char *, const char *) __attribute__((format_arg(2)));
int fpurge(FILE *);
void setbuffer(FILE *, char *, int __size);
int setlinebuf(FILE *);
int vasprintf(char * *restrict, const char * restrict, va_list) __attribute__((__format__ (__printf__, 2, 0)));





FILE *funopen(const void *,
     int (* _Nullable)(void *, char *, int __n),
     int (* _Nullable)(void *, const char *, int __n),
     fpos_t (* _Nullable)(void *, fpos_t, int),
     int (* _Nullable)(void *));
# 507 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/_stdio.h" 3 4
# 1 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/secure/_stdio.h" 1 3 4
# 32 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/secure/_stdio.h" 3 4
# 1 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/secure/_common.h" 1 3 4
# 33 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/secure/_stdio.h" 2 3 4





extern int __snprintf_chk (char * restrict , size_t __maxlen, int, size_t,
     const char * restrict, ...);
extern int __vsnprintf_chk (char * restrict , size_t __maxlen, int, size_t,
     const char * restrict, va_list);

extern int __sprintf_chk (char * restrict , int, size_t,
     const char * restrict, ...);
extern int __vsprintf_chk (char * restrict , int, size_t,
     const char * restrict, va_list);
# 508 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/_stdio.h" 2 3 4
# 62 "/Library/Developer/CommandLineTools/SDKs/MacOSX.sdk/usr/include/stdio.h" 2 3 4
# 2 "radius.c" 2


int main(void)
{
    int radius;
    float area;

    printf("Enter the radius: ");
    scanf("%d", &radius);

    area = 3.14 * radius * radius;
    printf("Area = %f\n", area);

    return 0;
```

This view shows the code after preprocessing. The contents required from stdio.h are inserted, and the macro PI in the calculation is replaced by its value 3.14 before compilation.

4. Viewing the ARM64 Assembly

```
	.cfi_offset w30, -8
	.cfi_offset w29, -16
	mov	w8, #0			; =0x0
	str	w8, [sp, #16]		; 4-byte Folded Spill
	stur	wzr, [x29, #-4]
	adrp	x0, l_.str@PAGE
	add	x0, x0, l_.str@PAGEOFF
	bl	_printf
	mov	x9, sp
	sub	x8, x29, #8
	str	x8, [x9]
	adrp	x0, l_.str.1@PAGE
	add	x0, x0, l_.str.1@PAGEOFF
	bl	_scanf
	ldur	s1, [x29, #-8]
					; implicit-def: $d0
	mov.16b v0, v1
	sshll.2d	v0, v0, #0
					; kill: def $d0 killed $d0 killed $q0
	scvtf	d1, d0
	mov	x8, #34079		; =0x851f
	movk	x8, #20971, lsl #16
	movk	x8, #7864, lsl #32
	movk	x8, #16393, lsl #48
	fmov	d0, x8
	fmul	d0, d0, d1
	ldur	s2, [x29, #-8]
					; implicit-def: $d1
	mov.16b v1, v2
	sshll.2d	v1, v1, #0
					; kill: def $d1 killed $d1 killed $q1
	scvtf	d1, d1
	fmul	d0, d0, d1
	fcvt	s0, d0
	stur	s0, [x29, #-12]
	ldur	s0, [x29, #-12]
	fcvt	d0, s0
	mov	x8, sp
	str	d0, [x8]
	adrp	x0, l_.str.2@PAGE
	add	x0, x0, l_.str.2@PAGEOFF
	bl	_printf
	ldr	w0, [sp, #16]		; 4-byte Folded Reload
	ldp	x29, x30, [sp, #32]	; 16-byte Folded Reload
	add	sp, sp, #48
	ret
	.cfi_endproc
					; -- End function
	.section	__TEXT,__cstring,cstring_literals
l_.str:					; @.str
	.asciz	"Enter the radius: "

l_.str.1:				; @.str.1
	.asciz	"%d"

l_.str.2:				; @.str.2
	.asciz	"Area = %f\n"
```

Clang has translated the C program into ARM64 assembly for the Mac. These instructions prepare the printf and scanf calls, perform the area calculation, store the result, and return from main.

5. Creating the Object File

The command clang -c radius.c -o radius.o creates machine code without linking. The file command identifies radius.o as a 64-bit Mach-O ARM64 object, confirming successful object-file generation.

```
        str     x8, [x9]
        adrp    x0, l_.str.1@PAGE
        add     x0, x0, l_.str.1@PAGEOFF
        bl      _scanf
        ldur    s1, [x29, #-8]
                                        ; implicit-def: $d0
        mov.16b v0, v1
        sshll.2d        v0, v0, #0
                                        ; kill: def $d0 killed $d0 killed $q0
        scvtf   d1, d0
        mov     x8, #34079                      ; =0x851f
        movk    x8, #20971, lsl #16
        movk    x8, #7864, lsl #32
        movk    x8, #16393, lsl #48
        fmov    d0, x8
        fmul    d0, d0, d1
        ldur    s2, [x29, #-8]
                                        ; implicit-def: $d1
        mov.16b v1, v2
        sshll.2d        v1, v1, #0
                                        ; kill: def $d1 killed $d1 killed $q1
        scvtf   d1, d1
        fmul    d0, d0, d1
        fcvt    s0, d0
        stur    s0, [x29, #-12]
        ldur    s0, [x29, #-12]
        fcvt    d0, s0
        mov     x8, sp
        str     d0, [x8]
        adrp    x0, l_.str.2@PAGE
        add     x0, x0, l_.str.2@PAGEOFF
        bl      _printf
        ldr     w0, [sp, #16]                   ; 4-byte Folded Reload
        ldp     x29, x30, [sp, #32]             ; 16-byte Folded Reload
        add     sp, sp, #48
        ret
        .cfi_endproc
                                        ; -- End function
        .section        __TEXT,__cstring,cstring_literals
l_.str:                                 ; @.str
        .asciz  "Enter the radius:"

l_.str.1:                               ; @.str.1
        .asciz  "%d"

l_.str.2:                               ; @.str.2
        .asciz  "Area = %f\n"

.subsections_via_symbols
bhavya@Bhavyas-MacBook-Air ~ % cc -save-temps radius.o
bhavya@Bhavyas-MacBook-Air ~ % cat radius.o
????
    ? 8???__text__TEXT???       ?__cstring__TEXT? ?__compact_unwind__LD? ?2     ?H
                                                                                  P????{????R???????????#?(??????_? ?N? ?a^祸?h=????(??ga??_?A?N!? !?aa@b?C??C_??"???????@??{B?????_?Enter the radius:%dArea =
%f
??-?L?=-4L0= -L=;?2?#?,?l_.str_main_printf_scanfltmp2l_.str.2ltmp1l_.str.1ltmp0%
bhavya@Bhavyas-MacBook-Air ~ % ls
a.out           Documents       Library         Music           Public          radius          radius.c        radius.o
Desktop         Downloads       Movies          Pictures        radi            radius.bc       radius.i        radius.s
bhavya@Bhavyas-MacBook-Air ~ %
```

6. Result

The terminal views show the complete flow from C source code to execution, preprocessing, ARM64 assembly, and object code. The experiment was completed successfully.